

Maria Eleni Zoumpoulidi

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I am an incoming Master's student and Researcher interested in LLMs' reasoning, interpretability and explainability.

EDUCATION

09/2026 - 09/2027 **Imperial College London**, London, UK

- Master of Research (MRes) in AI and Machine Learning

09/2019 - 03/2025 **National Technical University of Athens (NTUA)**, Athens, Greece

- Diploma (MEng) in Electrical and Computer Engineering (5-year Integrated Master's degree, 300 ECTS)
 - Grade: 7.98/10.00 "Very Good"
 - Relevant Coursework: Pattern Recognition (10/10), Speech & Natural Language Processing (9/10), AI (8/10), Neural Networks and Deep Learning (7/10), Computability & Complexity (10/10)
 - Diploma thesis: "BloomWise: Enhancing Problem-Solving capabilities of LLMs using Bloom's-Taxonomy-inspired prompts", [link](#) (supervisor: prof. Alexandros Potamianos ([Scholar](#))) (10/10)

RESEARCH AND PROFESSIONAL EXPERIENCE

03/2025 – 06/2026 **Associate Researcher**, Institute for Language and Speech Processing, Athena Research Center, Greece

- Conducted research on LLMs' reasoning, interpretability, and explainability.
- **Led and first-authored** *BloomWise* (improving LLMs' reasoning through cognitively-inspired in-context learning), *BloomXplain* (an explainability framework and benchmark), and a work on cognitive profiling of LLMs (ongoing);
- **Co-led and co-first-authored** two works on interpretability and faithfulness regarding mathematical solvability detection (ongoing).
- Performed LLMs' fine-tuning and inference on HPC systems.
- Developed retrieval-augmented generation (RAG) pipelines.

07/2022 - 08/2022 **Intelligent Decisions Intern**, Qualco (international fintech solutions provider), Greece

- Responsibilities: production of prototype explainers for NLP-based ML predictive models

FIRST AUTHOR PUBLICATIONS AND PREPRINTS

- **BloomWise: Enhancing Problem-Solving capabilities of Large Language Models using Bloom's Taxonomy-Inspired Prompts** ([paper](#))
[Maria-Eleni Zoumpoulidi](#), Georgios Paraskevopoulos, Alexandros Potamianos
MathNLP Workshop @ EMNLP 2025 (archival, oral presentation)
- **BloomXplain: A Framework and Benchmark Dataset for Pedagogically Sound LLM-Generated Explanations Based on Bloom's Taxonomy** ([paper](#))
[Maria-Eleni Zoumpoulidi](#), Eleni Batsi, Georgios Paraskevopoulos, Vassilis Katsouros, Alexandros Potamianos
LLM-eval Workshop @ NeurIPS 2025 (non-archival, poster)
- **Knowledge Knows, Verbalization Tells: Disentangling Latent Directions for Mathematical Solvability in LLMs** ([paper](#))
Nikolaos Xiros*, [Maria-Eleni Zoumpoulidi*](#), Georgios Paraskevopoulos, preprint ([co-first author, randomized author order](#))
Arxiv preprint

VOLUNTEERING EXPERIENCE

- Coordinator of the Machine Learning team, IEEE NTUA Student Branch, Computer Society Chapter, Apr 2023 - Oct 2023

EXTRACURRICULAR ACTIVITIES

- Erasmus+ Instructor: AI & Digital Transformation in Higher Education, Jun 2025
- GoCERN Summer School (CERN-NTUA-Arsakeio Tositseio School), Jun 2019
- Participation in the Athens Model of United Nations (MUN), Mar 2017

SKILLS

- Software: Python, C++, SQL, Bash, Git, Operating Systems
- Machine Learning: PyTorch, HuggingFace, NumPy, Pandas

LANGUAGES

- **English** (fluent-Proficiency), **French** (independent user-Delf B2), **Greek** (native)